



Real-Time Motor Imagery-Based Brain–Computer Interface System by Implementing a Frequency Band Selection

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Abstract

Motor imagery-based brain–computer interfaces (MI-BCIs) are a promise to revolutionize the way humans interact with machinery or software, performing actions by just thinking about them. Patients suffering from critical movement disabilities, such as amyotrophic lateral sclerosis (ALS) or tetraplegia, could use this technology to interact more independently with their surroundings. This paper aims to aid communities affected by these disorders with the development of a method that is capable of detecting the intention to execute movements in the upper extremities of the body. This will be done through signals acquired with an electroencephalogram (EEG), their conditioning and processing, and their subsequent classification with artificial intelligence models. In addition, a digital signal filter will be designed to keep the most characteristic frequency bands of each individual and increase accuracy significantly. After extracting discriminative statistical, frequential, and spatial features, it was possible to obtain an 88% accuracy on validation data with a random forest (RF) model when it came to detecting whether a participant was imagining a left-hand or a right-hand movement. Furthermore, a convolutional neural network (CNN) was used to distinguish if the participant was imagining a movement or not, which achieved 78% accuracy and 90% precision. These results will be verified by implementing a real-time simulation with the usage of a robotic arm.

Keywords Motor imagery-based brain–computer interface (MI-BCI) · Event-related desynchronization and synchronization (ERD/ERS) · Finite impulse response (FIR) · Common spatial patterns (CSP) · Short-time Fourier transform (STFT) · Real-time systems

1 Introduction

Spinal cord injuries (SCIs) occur when there is significant damage to the nerve pathways that interconnect the brain with peripheral nervous systems, impeding the correct transmission of neural impulses [1]. If the harm is severe, SCIs can produce a complete movement loss, resulting in paraplegia or tetraplegia. According to the World Health Organization (WHO), there are 250 thousand to 500 thousand of new patients worldwide suffering from SCIs yearly who often need constant assistance from familiars, friends, or caretakers [2]. These concerning numbers make it essential to seek a technology that could allow them to be more independent.

A brain–computer interface (BCI) is a system that could efficiently aid these patients, providing a direct communication pathway between the brain and a computer regardless of the presence of a SCI or a neurodegenerative disease such as amyotrophic lateral sclerosis (ALS) [3].

Various experiments have been implemented in order to prove the reliability and accuracy of BCIs when it comes to detecting the movement intentions of an individual, known as motor imagery BCI (MI-BCI). One relevant example is the company Neuralink, which has implanted an array of 1024 miniaturized electrodes into the motor cortex of a monkey and demonstrated that the animal was able to precisely play “Pong” by just imagining hand gestures [4]. However, implanted BCIs are nowadays risky for the user’s health since they require complex neurosurgery procedures.

As an alternative, an electroencephalogram (EEG) can be used since the electrodes are superficially attached to the person’s scalp. Nevertheless, non-intrusive BCIs present drawbacks that must be addressed for proper system operation. First and foremost, the signals captured from an EEG

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have an amplitude of microvolts, which makes it highly noise-susceptible, especially if there are no active electrodes [5]. Secondly, it is challenging to find the source of the neural activity if there is not a sufficient number of electrodes [6]. Thirdly, owing to the poor conductivity of the skull, there is a substantial spatial resolution loss (spatial smearing effect) [7]. Lastly, since the electrodes are manually placed, their position and quantity of hydrogel used can slightly vary from session to session, which can affect the results of the BCI [6]. Sophisticated mathematical techniques must be carried out to solve these complications, thus considerably increasing the performance of the system.

Currently, most MI-BCI systems only support synchronization control, where users are prompted by the system to imagine at specific times but cannot stop the task at their discretion. This undermines the autonomy of the MI-BCI system. Asynchronous mode, on the other hand, allows users to control the pace of the BCI system themselves. The challenge in achieving asynchronous control is accurately distinguishing when the user starts or stops imagining tasks, i.e., the resting state. Discerning between the rest and the task state proves to be an arduous task since there is no clear demarcation between them. Few studies have been published mentioning the detection of the resting state. For instance, T. Shiratori et al. [34] classified the three classes (both hand imageries and the resting) with an accuracy of 56.7% with the use of multi-class common spatial filter (mCSP), which should be significantly improved. M. Tavakolan et al. [35] could detect the resting state with an average accuracy of 75.8% by using variants of the CSP, but a unique classifier was used thus affecting the accuracy of the other tasks and the overall precision of the BCI, prompting to undesired false positives.

The main purpose of this paper is to implement an asynchronous noninvasive MI-BCI and design algorithms and methods that will be capable to distinguish whether the user has the intention to move the right or the left hand (just by thinking of it). To do so, two classifiers must be defined, the first one to detect if the user is in the resting state or not, and, if the resting state is not detected, the second classifier decides on the type of MI (right or left-hand imagery). EEG signals must be understood in the context of MI to painstakingly choose which signal processing procedures, such as common spatial patterns (CSP), are more suitable for better extraction of pertinent features. These characteristics will act as inputs to train various artificial intelligence (AI) models (i.e., convolutional neural networks (CNN), random forests (RF), support vector machines (SVMs), and linear discriminant analysis (LDA)) that will be evaluated with certain metrics to discriminate which one performs the best. An appropriate tuning and optimization of AI models are of great importance in this work. Compared to other works, this paper proposes a novel methodology to detect accurately the

resting state, while the EEG is continuously sending data. This is a paradigm that has been largely overlooked in other MI-BCI studies, as most researchers in the field mainly focus in right and left-hand classification or other multiclass MI. Therefore, the system designed can perform accurate asynchronous real-time classification to control a robotic arm with the help of a short-time Fourier transform (STFT) analysis.

In parallel, finite impulse response (FIR) filters will be carefully studied and analyzed to exclude insignificant spectral information since the CSP are proven to behave poorly without proper filtration [8]. Hence, this project furthermore intends to design a digital filter in a real-time operation with the help of event-related desynchronization and synchronization (ERD/ERS) maps, considering the pros and cons when working with higher- or lower-order filters.

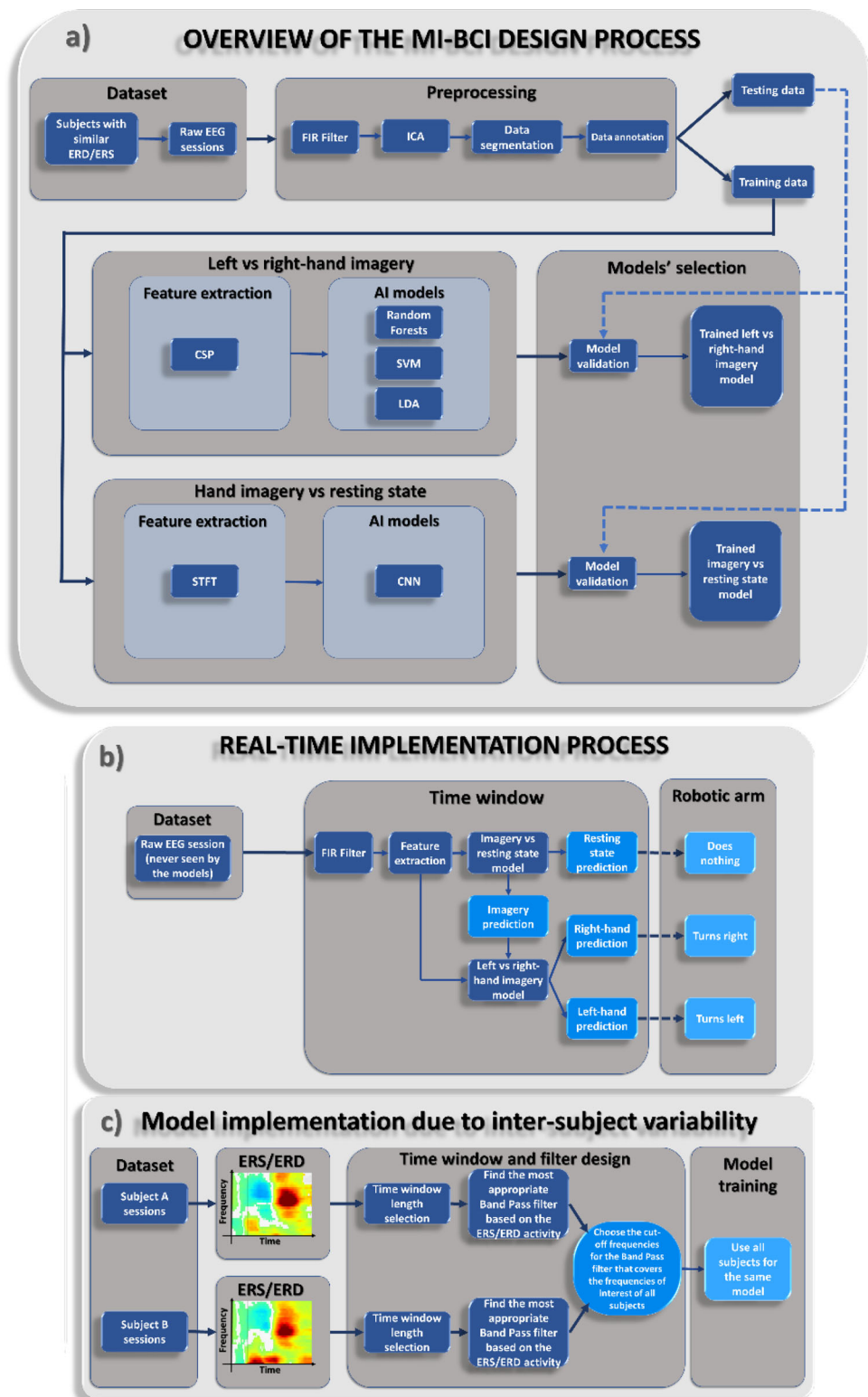
2 Materials and Methods

One of the most extensive publicly available MI datasets [9] was used for designing, training, and evaluating the MI-BCI classifier. The dataset contains EEG recording sessions where different participants had to imagine left and right-hand movements. Each subject's data is analyzed to understand the signals, visualizing the effect of ERS and ERD in a spectrogram. By seeing the most relevant frequency bands from each subject, the signals were bandpass filtered digitally to exclude unnecessary information that may disturb the artificial intelligence (AI) models. Subsequently, the data were prepared for training the classifiers, including artifact removal (eye-blinking) and feature extraction with advanced methods, i.e., CSP and STFT.

From there, two AI models were designed; the first one to classify the hands' imagery (left and right-hand) (Model LR), and the second to distinguish between an imagery event and the resting state of the user (Model IR). It must be highlighted that the AI models will be trained with data from multiple subjects; hence, the inter-subject variability in the ERS/ERD frequency bands must be taken into account. Before choosing the cutoff frequencies and the window's length that will be used for the model, each participant must undergo a calibration process to understand his frequential information in order to implement a broader bandpass filter that covers the essential spectral behavior of all the subjects collectively (see Fig. 1a and c).

Once the models were trained and evaluated, the MI-BCI was implemented on an hour MI session (from the dataset) unseen by the AI models. That was done by coding a program that simulates a real-time signal acquisition, where the signals of the session are introduced inside a time window the same way that they would do if the data were obtained live from an EEG. Essentially, this means that the simulated MI session signals were introduced into the AI models in the

Fig. 1 **a** Flow diagram overviewing the design process of the MI-BCI; **b** Flow diagram detailing how to implement the MI-BCI in real-time. **c** Diagram explaining how to correctly design the frequential filter and the time window size



same way that EEG signals would be acquired in a real-time MI session. With this approach, it is possible to evaluate the AI models' performance in a scenario that closely resembles a real-world MI-BCI application, where the signals are recorded in real time from an EEG. Afterward, the signals are filtered, and the features are extracted. Notice that this real-time execution requires a proper digital filter design, avoiding significant delay between the mental imagery and its detection. If the length of the FIR filter is too long (in order to achieve greater selectivity), the delay can even reach several seconds, which can significantly impact the smoothness and latency of the system. Next, the correspondent features are introduced to Model IR. If the prediction is a resting state, no action will be taken. Otherwise, if it is an imagery prediction, the appropriate features will be introduced to Model LR, which will classify if the signal present in the time window is a left- or right-hand imagery (see Fig. 1b). By doing so, it is possible to control a robotic arm directly from the mental activity of the subject.

2.1 The Dataset

The database contains four different paradigms; however, this paper will only work with one of them: the classical (CLA). It consists of the imagery of left and right-hand movements and the resting state, including data from six different subjects and a total of fourteen sessions with approximately 1 h each. All the participants were healthy students without any psychiatric disorder diagnosed; they are nicknamed "Subject X," being X a letter from A to F. The EEG used was the EEG-1200 JE-921A system from the company Nihon Kohden in the international 10–20 configuration.

2.2 Common Average Referencing (CAR)

Many researchers recommend re-referencing all the channels to the average value of all the electrodes (common average referencing (CAR)) [10, 11]. This technique reduces considerably noisy sources that may be problematic. It also acts as a spatial filter by reducing the presence of shared activity between electrodes [12]. Therefore, this method was implemented to improve the signals' quality by applying the following equation:

$$x_i^{\text{CAR}}(t) = x_i(t) - \frac{1}{C} \sum_{j=1}^C x_j(t)(V), \quad (1)$$

where C is the number of the total electrodes used, x_i is the value of the electrode i , x_i^{CAR} is the value of the electrode i after the CAR, and x_j is the value of the electrode j .

2.3 Event-Related Desynchronization and Synchronization

Many researchers have proven that MI tasks can stimulate the motor cortex in similar ways as an actual motor execution, producing brainwave oscillations known as sensorimotor rhythms (SMRs) [13, 14].

SMRs have two frequency bands that must be considered: the Mu band (8–13 Hz), which reflects alpha fluctuations located in the sensorimotor region of the brain, and the Beta band (13–30 Hz). Both have shown that they are relevant for identifying the activity of neural groups while imagining or performing a movement [15]. The spectral information of these bands is essential for the MI-BCI system to perceive event-related synchronizations (ERS), occurring when neurons produce oscillations that increase in amplitude after the imagery task, and event-related desynchronizations (ERD), in which there is an amplitude decrease after the MI. As their name suggests, ERS and ERD are event-related potentials (ERP), defined as a cerebral response in front of a produced event, in this case, the MI.

Plotting the ERD and ERS maps is extremely helpful to understand the frequential behavior of the left and right-hand imagery over time. With this information, the relevant frequency bands can be detected in order to posteriorly design the cutoff frequencies of the band-pass filter, attenuating insignificant bands from the spectrum for better performance.

Furthermore, it is possible to check the most recurrent reaction time of each participant after the instructor's indications to perform imageries. In addition, the time window length is estimated based on the time in which the ERD/ERS activity lasted on the map. These procedures are necessary to later segment the data for training the AI models.

The ERD/ERS maps were drawn following instructions described in [16–18]. They are computed by clustering and averaging the time–frequency representations of all the periods that contain an imagery event. These periods start one second before the instruction to perform the MI task (represented as a dashed line) and end four seconds later. It is important to remember that, as most of the MI events occur in the motor cortex, the ERD/ERS maps will be plotted in the central electrodes of the EEG (C3, C4, and Cz).

The blue tonalities of the maps represent a low activity (normally an ERD) and the red ones have a higher activity (usually an ERS). The darker the color is, the more mental responsiveness there is. Therefore, it is optimal to detect ERS and ERD with this property as it means that the participant has an outstanding ability to control the MI tasks.

As a demonstration of how to analyze the ERD/ERS maps, Fig. 2 displays the ERD/ERS patterns for Subject C. For the left-hand imagery (upper figures), there is a dominant activity in channel C4, where an ERD in the Beta band (17–26 Hz, 0.1–1 s) is followed by an ERS in the Beta (15–21 Hz,

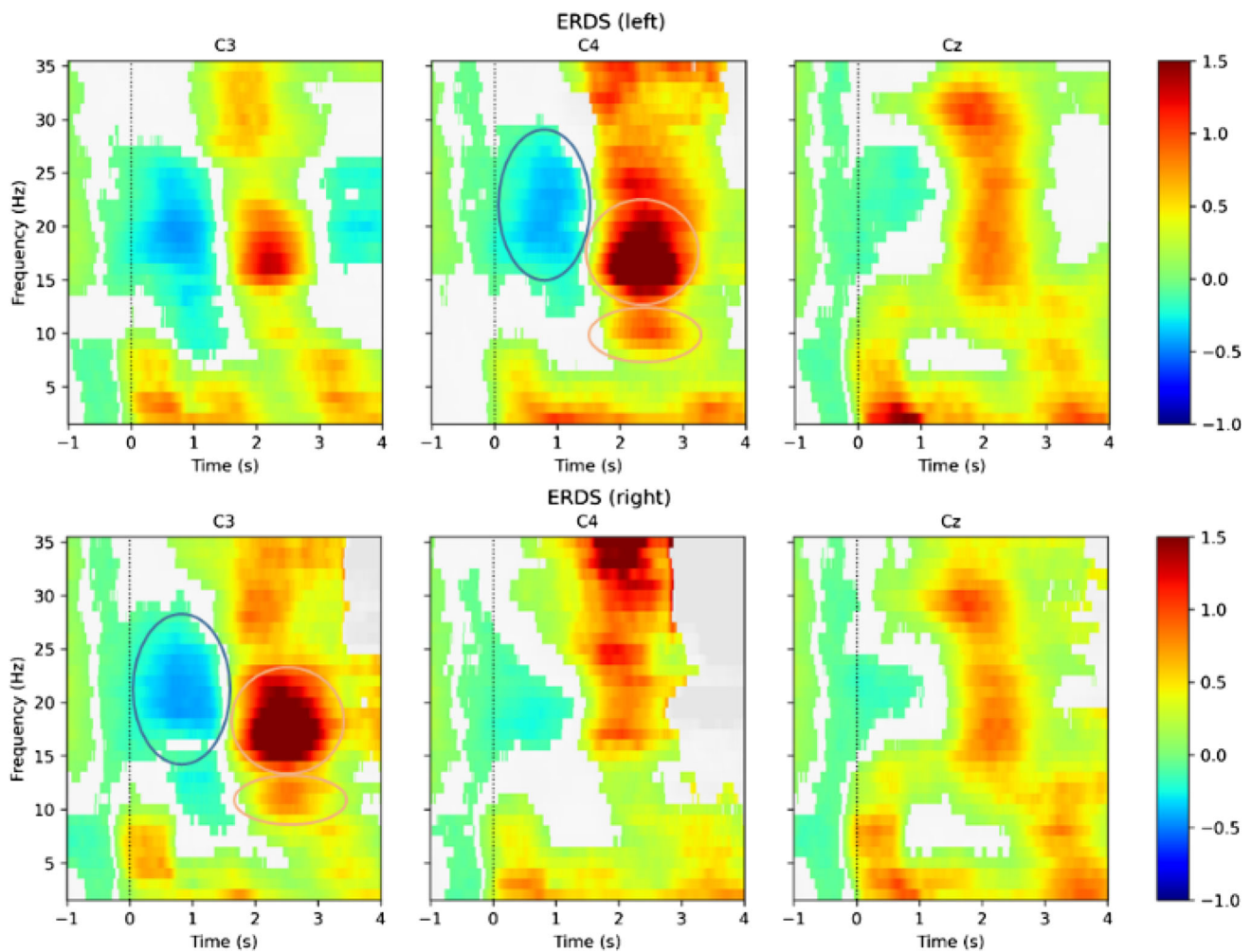


Fig. 2 Subject C's brain activity (ERD/ERS) in channels C3, C4, and Cz while imagining left and right-hand movements. X-axis is time, Y-axis is frequency, and colors represent mental activity (blue = low, red = high). Dashed line indicates imagery task, and circles are areas of interest

1.8–2.8 s) and the Mu band (10–13 Hz, 2–3 s). Notice that a similar pattern occurs in electrode C3 but with much less power. For the right hand (lower figures), channel C3 shows the same pattern as the C4 electrode of the left-hand imagery, with ERDs and ERSs in the same frequency bands and times.

After analyzing all the ERDS maps of all the subjects, many aspects must be pointed out before segmenting the data and training the BCI's AI models.

First and foremost, from all the dataset's subjects, three participants were discarded from the study as they had abnormal behaviors that could preclude the distinction between the left and right-hand imagery. That could be caused due to the lack of experience while carrying out MI tasks, which could be mastered by practicing this mental chore for hours or with neurofeedback [19, 20]. The only subjects with fair-quality recordings are subjects *C*, *E*, and *F*.

Secondly, only Subject *C* presented ERD, while the remaining participants jumped from a neutral state to an ERS directly. Hence, from all the time–frequency representations,

only the ERS will be considered for training the AI models. Furthermore, it has been seen that the activity in the Mu band was not present in all the subjects, and, when it did, its intensity was noticeably lower than the ERS appearing in the Beta band. That is to say that the band-pass filter that will be implemented should only maintain the ERS of the Beta band. Moreover, the time window length used for segmenting the data should be sufficient to cover the ERS activity so that the model can process it correctly.

Finally, it is important to note that ERS and ERD may appear in a slightly different bandwidth depending on the subject [21]; this complicates the implementation of a generic MI-BCI. Hence, for the correct functioning of the system, the inter-subject variability must be assessed in order to know how to train and implement the BCI. As shown in Fig. 1c, after obtaining the ERS/ERD maps, the researcher must decide which time window length and cutoff frequencies should be used for each participant to highlight the ERS. Subsequently, all the subjects will be compared to check if



Table 1 Inter-subject variability evaluation

Subject	Time window length	Frequency band (Hz)	Frequency band of the final model (Hz)
C	0.9 s	15–21	15–26
E	0.9 s	18–24	
F	0.9 s	20–26	

their ERS have similarities. Table 1 compares the selected subjects to determine the most appropriate frequency band for the final model; subjects *C*, *E*, and *F* will be trained with a window length of 0.9 s and a bandpass filter with lower and higher cutoff frequencies of 15 Hz and 26 Hz, respectively.

2.4 Digital Filtering

After deciding which frequency bands are suitable, a band-pass digital filter must be designed to remove undesired frequencies; this is an important step in this MI-BCI design as a well-designed filter can eradicate noise, interference, artifacts, or mental activities that are not related to MI.

Finite impulse response (FIR) filters have clear benefits that prevail over the IIR filters, having a linear phase and avoiding instability scenarios is of great interest while implementing a bandpass filter for biomedical purposes, especially in MI-BCI applications.

Some considerations must be taken into account before designing the FIR filter:

- Filter type** In this BCI application, the filter will be a bandpass, having an upper and a lower cutoff frequency. Additionally, it must be decided which passband ripple is accepted. Moreover, one must consider which transition band is desired. Remember that the narrower the transition band, the more selective the filter is. Lastly, it is important to ensure that the stopband ripple does not considerably affect the output.
- Window type** For avoiding abrupt changes in the tails of the filter's impulse response, it is preferable to operate with a window function. The Bode of an FIR filter presents a principal lobe (which is the most important) and many undesired sidelobes. By smoothing the extremes of the impulse response with a window (Hamming, Blackman, Hann, etc.), the sidelobes attenuate even more, which is optimal. The trade-off is that the principal lobe width augments and filter selectivity can be affected. To solve this problem, one can increase the window's length (see next point).
- Window length** The window length defines the number of lobes present in the Bode as well as the filter's order.

Table 2 Primary lobe width and the peak of the secondary lobes of the different windows types

Window type	Approximate width of main lobe	Peak sidelobe amplitude (dB)
Rectangular	$\frac{4 \cdot \pi}{M+1}$	– 13
Bartlett	$\frac{8 \cdot \pi}{M}$	– 25
Hanning	$\frac{8 \cdot \pi}{M}$	– 31
Hamming	$\frac{8 \cdot \pi}{M}$	– 41
Blackman	$\frac{12 \cdot \pi}{M}$	– 57

Remember that the higher the order, the more selective the filter is. However, the filter's order must not be too large since the delay will increase. One way of calculating this parameter is by looking at Table 2 and establishing the width of the first lobe as the normalized passband. For instance, if the cutoff frequencies are 15 Hz and 26 Hz, and the sampling frequency is 200 Hz, the normalized passband can be calculated as:

$$\begin{aligned} \text{PB}_n &= w_{2n} - w_{1n} = 2 \cdot \pi \cdot \frac{f_2 - f_1}{f_s} \\ &= 2 \cdot \pi \cdot \frac{26 - 15}{200} = 0.11 \cdot \pi \frac{\text{rad}}{s} \end{aligned} \quad (2)$$

If a Hamming window is used, it is possible to calculate the minimum length of the filter:

$$\text{PB}_n = \frac{8 \cdot \pi}{M} \rightarrow M = \frac{8 \cdot \pi}{\text{PB}_n} = \frac{8 \cdot \pi}{0.11 \cdot \pi} \approx 73 \text{ samples} \quad (3)$$

Thus, the length must be at least 73 samples, for example, 150 samples. The delay is then calculated, which is more than acceptable for a MI-BCI:

$$\text{Delay} = \frac{M - 1}{2 \cdot f_s} = \frac{150 - 1}{2 \cdot 200} = 372.5 \text{ ms} \approx 75 \text{ samples} \quad (4)$$

2.5 Independent Component Analysis (ICA)

As its name indicates, the independent component analysis (ICA) is a preprocessing tool capable of detecting and separating a defined number of statistically independent elements that composes a signal without having prior knowledge of the signal's content (blind source separation) [22]. As the EEG contains multiple electrodes that are distributed strategically around the scalp, the sources of these components can be localized. This technique is suitable for detecting and eliminating artifacts, such as the eye-blinks (Fig. 3) or the electromyographic activity of facial muscles, without altering the valuable signal [6]. The ICA seeks a linear

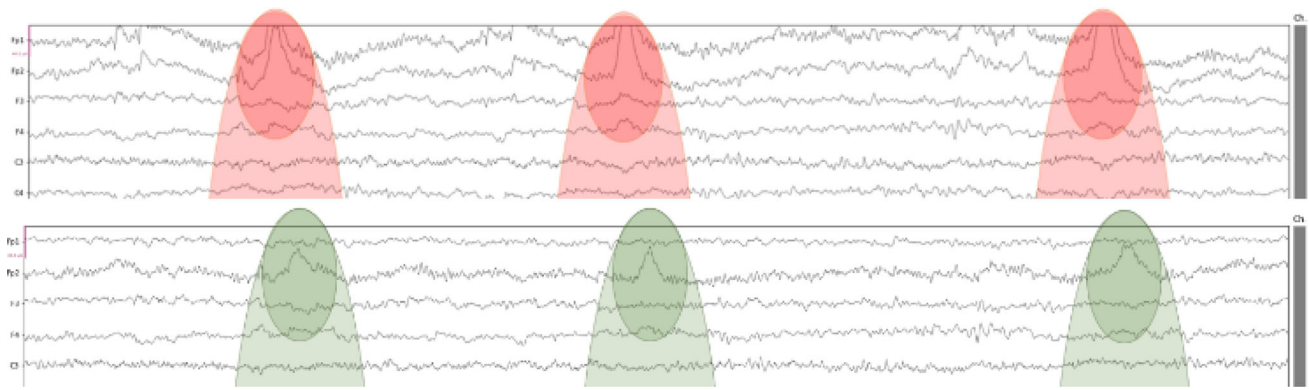


Fig. 3 Effect of applying the independent component analysis (ICA) to remove eye-blinking artifacts in the frontal electrodes. Red: before ICA. Green: after ICA

transformation to reduce as much as possible the statistical dependence of the signal's components [23–25].

2.6 Data Segmentation and Annotation

While the participants were carrying out the trials, they constantly followed the indications of the instructor, which saved all the taken actions inside a marker vector. One might think that the mental activity starts right after the marker; however, that is not the case, as there is a reaction time from the subject that introduces a delay between the indication and the actual MI task. This retard can be estimated by looking at the ERS/ERD maps when the Beta band ERS begins. After segmenting all the data, it is important to label each epoch to train and test the supervised AI.

2.7 Common Spatial Patterns (CSP)

The common spatial patterns (CSP) technique is one of the most practical feature extraction methods for MI applications, enabling a more effective differentiation between two classes, such as the left and right-hand imagery. With the CSP algorithm, it is possible to find a set of spatial filters that will highlight or de-emphasize the activity of various areas of the brain in order to maximize the variance of one class while minimizing the other one's variance. Consequently, both categories are more distinguishable [26].

Fundamentally, the CSP can be obtained after the diagonalization of two covariance matrices, one for each hand imagery [27–29]. This algorithm must be trained with labeled epochs (segmented and annotated trials) to find the optimal spatial filters' weights with the highest inter-class variance. Therefore, it is a supervised technique that applies a transformation to the data to find the most suitable spatial features of the signals [30]. Figure 4 illustrates the CSP gotten from Subject C, showing how each behaves on a topographic map. Notice that CSP0 and CSP3 present emphasized patterns in

electrodes C4 and C3, respectively; this is a clear indicator that the CSP algorithm can detect left and right-hand imagery correctly.

2.8 Short-Time Fourier Transform (STFT)

One of the AI models proposed involves using a convolutional neural network (CNN), which is usually trained with images; therefore, the EEG signals must undergo a process to describe their contents in a two-dimensional manner. For doing this, the spectrogram of the signals can be computed, a processing tool that enables a time–frequency analysis of the acquired data.

The STFT is employed to compute the spectrograph. This mathematical technique divides the signal into multiple time segments with the help of a shifting time window, which is usually a Hamming to avoid spectral leakage and preserve continuousness. Furthermore, an ensuing window does not begin from the final position of the current window as there is an overlap to not lose the spectral information due to the Hamming window. After having all the segmented signals, a discrete Fourier transform (DFT) is calculated for each of them; as a result, the frequential information is obtained over time [31, 32].

There is a trade-off that must be addressed regarding the resolution of the spectrum. The larger the window length used in the STFT is, the better the frequential resolution and the more flawed the temporal resolution is. The contrary effect happens when the window gets shorter. Consequently, one must choose wisely the length of the window and the amount of overlap allowed [33]. To select the optimal window length and overlap, a method that has proven effective is to average and combine spectrograms of multiple samples from the same subject and type of imagery to have a more accurate representation of the brain activity being measured. It is essential to repeat this process for different window lengths and overlaps to determine the optimal values. By comparing



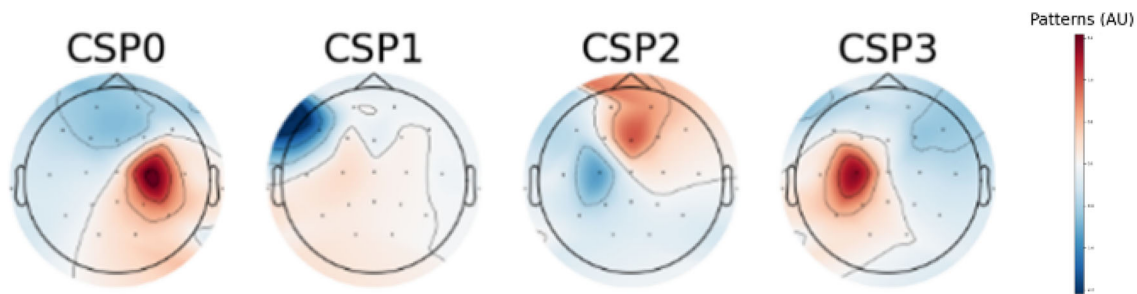


Fig. 4 Common spatial patterns (CSP) for Subject C. The color bar indicates the weights of the patterns while applying the filters. Red: positive coefficients. Blue: negative coefficients

and analyzing the resulting spectrograms, any flawed ones can be identified and discarded. Flawed spectrograms are often highly blurry or smeared in either frequency or time, making them easy to detect visually. In cases of uncertainty, multiple CNNs can be trained with different overlaps and window lengths, and the parameters of the most accurate result can be retained. The following equation describes how to carry out a STFT:

$$S(m, k) = \sum_{n=0}^{N-1} s(n + m \cdot N') \cdot w(n) \cdot e^{-j \cdot \frac{2\pi}{N} \cdot n \cdot k}, \quad (5)$$

where k is a number going from 0 to $N-1$, $S(m, k)$ specifies the time–frequency spectrograms for an index m , N is the window length, N' is the window's shifting step, and $w(n)$ involves the window function used. As the coefficients obtained from the previous equation are complex numbers, the magnitude of $S(m, k)$ must be calculated to obtain the spectrogram.

However, performing this operation for all the electrodes can be computationally expensive and could not allow a real-time analysis. Hence, only the central EEG electrodes (C3, C4, and Cz) will be chosen, as they are the most relevant for capturing the MI.

2.9 Artificial Intelligence Models

The last step for implementing the MI-BCI is to train AI models that will classify the three classes (left, right, and rest) accurately and precisely. The end goal is to find two different AI architectures, one for distinguishing between left and right-hand imageries and the other for differentiating imageries from the resting state. For doing so, manifold machine and deep learning methods will be tested to see which are the most suitable.

From all the segmented data, 80% will be used to train the models and the remaining 20% to test them; this ratio will ensure a correct evaluation of the models. It is important to clarify that the data will be shuffled randomly. For

the machine learning architectures, a K-fold Cross-validation will be used to have a more accurate estimate of the model's prediction performance.

The left and right-hand imagery classification (Model LR) will be tested mainly with machine learning methods, with the CSP as a feature extractor. The algorithms used are linear discriminant analysis (LDA), support vector machines (SVMs), and random forest (RF).

EEG-based BCI tasks have demonstrated the effectiveness of LDA, a linear classifier that seeks to project features onto a lower-dimensional space that maximizes between-class scatter and minimizes within-class scatter. Its simplicity and ability to reduce data dimensionality make it a valuable approach [36]. SVM is another popular classifier in BCI, known for its ability to handle high-dimensional data, robustness against overfitting, and strong generalization performance. By finding a hyperplane that maximally separates different classes, SVMs have shown to be effective in BCI [37]. RF is an ensemble of decision trees that aims to identify the most discriminative features to classify data into different classes. This method has demonstrated strong results in BCI by capturing nonlinear relationships between features and the target variable, as well as interactions between different features [38].

The imagery and resting-state classification (Model IR) will be performed with the CNN architecture, with the features obtained from the STFT. CNNs are a type of neural network that can automatically learn and extract features from the input data. They have shown great success in EEG-based BCI tasks, especially when applied to STFT of the EEG data. By analyzing different frequency bands, CNNs can capture the temporal and spectral characteristics of the EEG signals, leading to improved classification accuracy. Additionally, CNNs are robust to noise and can handle nonlinear relationships between the features and the target variable, making them a powerful tool for BCI applications [39].

The model's hyperparameters were selected with the help of appropriate tuning techniques to enhance the classifier's performance. This optimization is achieved with tools

available in the AI libraries of Python (Scikit-learn and TensorFlow), such as the Random Grid Search and the Bayesian Optimization; with them, one can assess the effects of choosing different hyperparameters. The hyperparameter used for the machine learning models is the following:

- *LDA* A singular value decomposition (SVD) solver and no shrinkage.
- *SVM* A radial basis function (RBF) kernel, a gamma of 0.00001, and a C parameter of 10,000.
- *RF* 100 number of estimators, a Gini criterion for measuring the quality of a split, a maximum depth of 11, a minimum number of samples to split an internal node of 5, a minimum number of samples to be considered as a leaf node of 4, and square root (sqrt) function to calculate the maximum number of features.

The imagery and resting-state classification (Model IR) will be performed with the CNN architecture, with the features obtained from the STFT.

The structure of the CNN is firstly composed of a 2D convolutional layer with 120 filters, a 3×3 kernel, a ReLU activation function, and an input size of (3,91,181); this is followed by a MaxPooling layer of size two and a dropout of 20%. The second convolutional layer has 240 filters, a 3×3 kernel, and a ReLU activation function, followed by a MaxPooling of size one and a dropout of 20%.

Afterward, a flatten layer prepares the data for entering inside two dense layers with 164 and 128 neurons, respectively. Both have a dropout of 20% and a ReLU activation function. The model ends with a one-neuron layer with a Sigmoid activation function. Regarding the CNN's hyperparameters, there is a learning rate of 0.0001 with a decay rate of 0.0001/300 and a momentum of 0.8; these parameters are introduced inside a stochastic gradient descent (SGD) optimizer. The loss will be evaluated with binary cross-entropy.

3 Results

After training all the models, the performance of the AI models must be evaluated to choose the most appropriate ones for real-time implementation. It is important to remark that each model was trained by using the data of Subjects *C*, *E*, and *F*. Bear in mind that the right vs left imagery models will be assessed in two separate ways:

- Mixing all the testing data of each subject for obtaining a general estimation of the models.
- Treating the inter-subject testing data independently to see how the general models perform for each participant.

3.1 Right vs Left-Hand Imagery (Model LR)

Based on the results presented in Table 3, it is evident that there is a difference in the accuracies to distinguish between left and right-hand imagery. While the overall accuracy of all three models—LDA, SVM, and RF—is similar, ranging from 82 to 83%, subject-specific analysis revealed distinct performance differences among them.

Subject C achieved higher accuracies in all three models, with an overall accuracy of 88% using RF. On the other hand, Subject E had slightly lower accuracies with an overall of 79% using all three models.

Considering the performance of the three models, the RF architecture demonstrated slightly better results than the other models, with an overall accuracy of 83%, reaching 88% in Subject C. Since the results are similar and potential hyperparameter's changes may cause that one model outperforms the other, the chosen model is the RF because it can reduce overfitting and improve the generalization performance of the model. Therefore, the RF model is the most suitable choice for distinguishing between left and right-hand imagery.

3.2 Imagery vs Non-Imagery (Model IR)

It is extremely important to avoid false positives so that a MI is not wrongly detected. Hence, accuracy and precision should be assessed conjunctively to find an optimal solution. The CNN has a 78% accuracy and a precision of 90%, which is reasonable for the BCI.

3.3 Real-Time Implementation

Once the best models are chosen, they will be used to classify an entire EEG session that was unseen while training or testing (offline) them. Therefore, the models will simulate an online categorization of the data, mimicking the behavior that would be obtained from a live collection of EEG data.

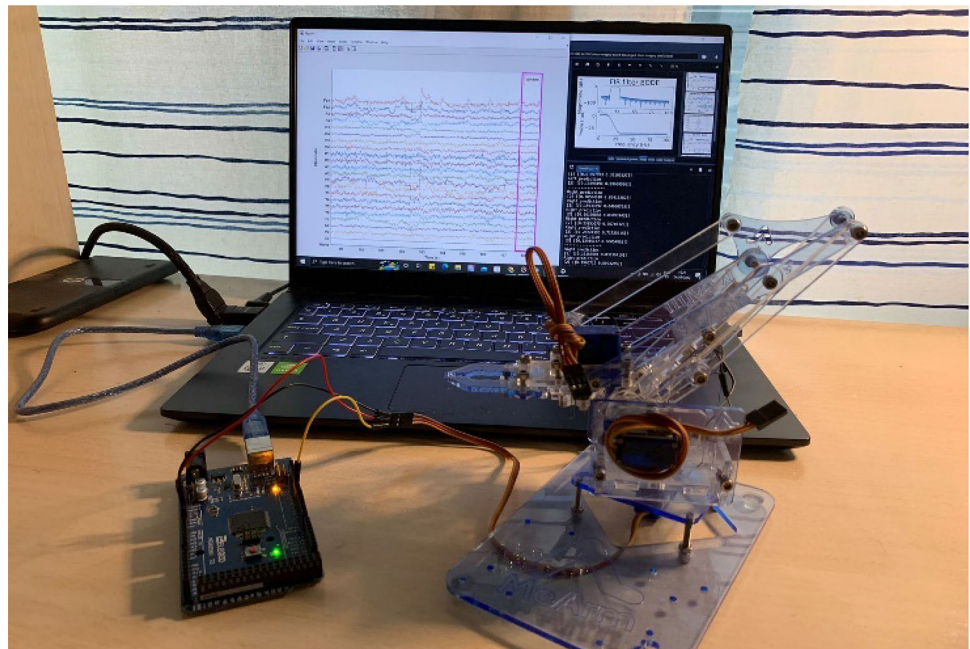
To carry it out, the EEG session will be introduced sequentially inside a window with a size of 180 samples (0.9 s since the sampling rate is 200 Hz). The data inside the window are filtered with the designed FIR filter; bear in mind that there will be a delay of 75 samples (justified in the digital filtering section). Subsequently, the STFT features are extracted, and the CNN model will classify whether the segment is an imagery or a resting state. If an imagery is detected, the CSP are obtained, and the RF model will distinguish between left and right-hand imagery. If a resting condition is detected, the BCI will ignore the segment (see Fig. 1b). Afterward, the data are shifted to 20 samples (0.1 s) into the window, and the cycle starts again.

In order to avoid noisy classifications, the model must predict four consecutive right or left-hand imagery to efficiently decide if a movement is really performed or not. For



Table 3 Accuracy and precision of the AI models

			General (All subjects)				Subject C			Subject E			Subject F		
			LDA	SVM	RF	CNN	LDA	SVM	RF	LDA	SVM	RF	LDA	SVM	RF
Right vs Left-hand imagery	CSP	Accuracy (%)	82	82	83	–	87	88	88	78	79	79	80	79	81
		Precision (%)	87	88	87	–	92	93	94	87	87	84	84	83	83
Imagery vs non-imagery	STFT	Accuracy (%)	–	–	–	78									
		Precision (%)	–	–	–	90									

Fig. 5 The setup of the MeArm robotic manipulator

instance, if the model has been predicting a resting state for 10 s and at 10.1 s detects a right-hand movement continued by a passive state, this would be considered a noisy classification.

The last step is to move a MeArm robotic manipulator left or right, depending on the prediction; this is achieved by establishing serial communication between Python and Arduino Mega 2560. In other words, a servomotor is activated based on the model's outputs. The setup is shown in Fig. 5.

A set of images (Figs. 6, 7, 8, 9, 10 and 11) will highlight some of the real-time predictions from an unseen session, showing cases in which the models perform flawlessly and others where there are some faults (see the captions of each illustration to have a detailed explanation). On the one hand, the red and green color bands indicate the instructor's indication to imagine a right or left-hand movement, respectively. On the flip side, the orange and blue color bands denote the

actual right and left-hand imageries predictions, correspondingly. The white regions also indicate that a resting state was predicted. Notice that there will always be a short delay between the indication and the prediction; this is caused due to the FIR filter and the subject's response time after seeing the stimuli.

Figures 6 and 7 indicate that all the classes were correctly classified, and the resting state was successfully detected. Figures 8 and 9 show that almost all the classes were correctly classified, and the resting state was successfully detected. However, there are cases where right imageries get detected as left, left imageries detected as right, and some imageries that were detected as resting. Figure 10 shows how well the model predicts a resting state, being passive when there is no indication to perform left or right-hand imageries. Figure 11 illustrates that sometimes an imagery state is detected even though the patient is in a resting state. These cases are not

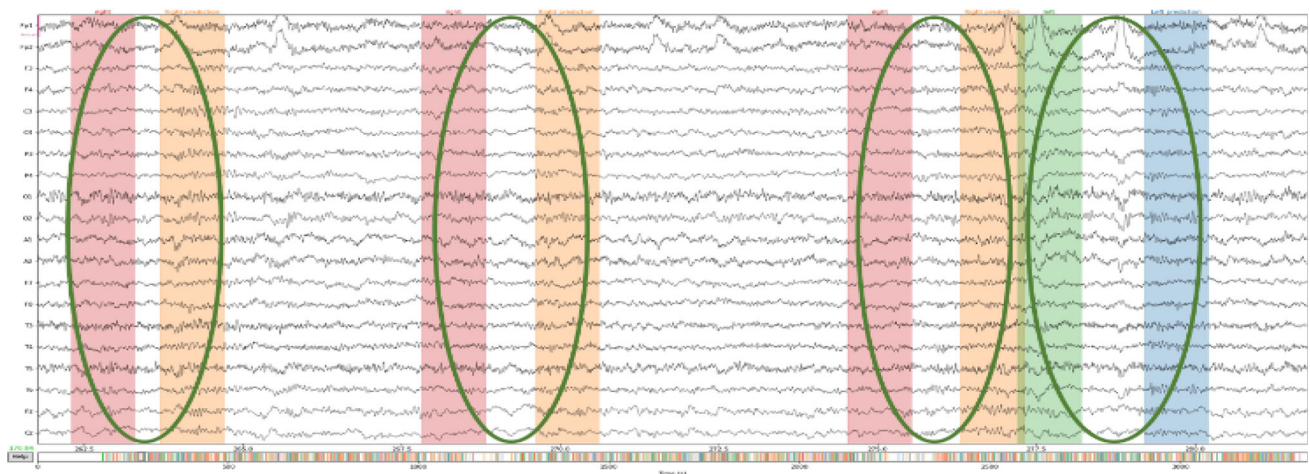


Fig. 6 Predictions on an unseen session. All the classes were correctly classified (green circles), and the resting state was successfully detected. Red band: right imagery indication. Green band: Left imagery indication. Orange band: Right prediction. Blue band: Left prediction

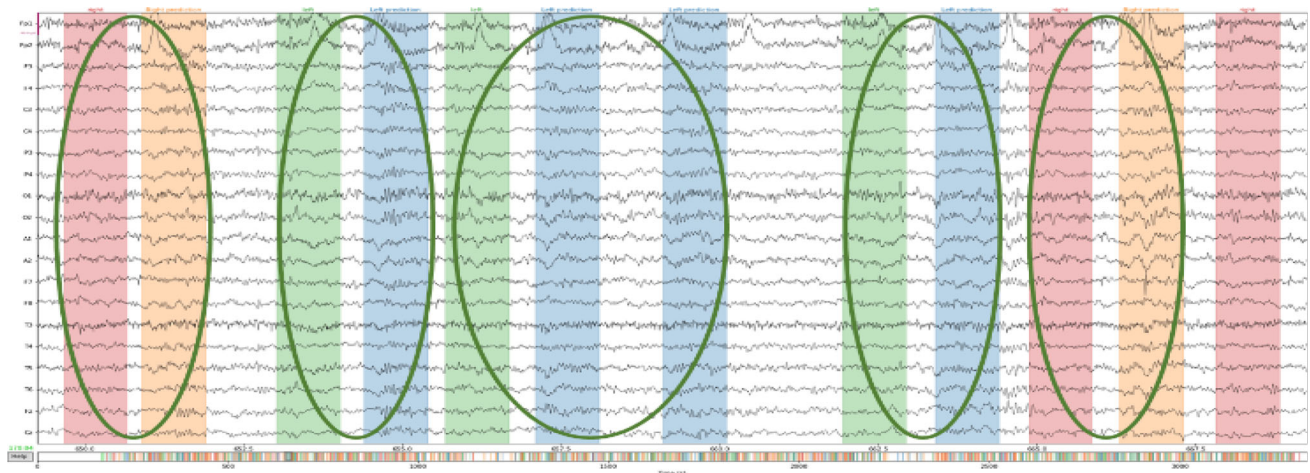


Fig. 7 Predictions on an unseen session. All the classes were correctly classified (green circles), and the resting state was successfully detected. Red band: right imagery indication. Green band: Left imagery indication. Orange band: Right prediction. Blue band: Left prediction

abundant since the precision of the imagery vs non-imagery model is high.

The imagery vs resting state model performed well, accurately predicting the resting state class 93% of the times. This high accuracy is due to the use of the robust algorithm that filters out noisy classifications mentioned before. However, the model also classified 21% of the imagery samples as resting. This trade-off between sensitivity and specificity was intentional, as it is preferable to have fewer false positives even if some imagery samples are missed. In the right vs left-hand imagery model, the model correctly predicted 84% of the classes. Overall, as expected from the results of the offline classification, the BCI system detects the hand imageries and the resting state precisely, outputting most of the time the correct category.

4 Discussion

In this work, we successfully implemented a BCI system that can translate EEG signals containing the imaginations of hand movements coming from the motor cortex activity into commands for controlling a robotic manipulator.

The first step taken was to understand the data to decide on further actions to consider. To reduce the spatial smearing effect, the common average referencing has been shown to better localize the sources of the EEG signals.

Furthermore, the ERD/ERS maps were extremely useful for comprehending the frequential behavior of each imagery over time. From the six subjects analyzed, only Subject C had a map that mostly reassembled the theoretical one, having clear ERD followed by an ERS in the Beta band and an ERS in the Mu band. To our surprise, the rest of the subjects lacked any ERD. Moreover, most participants only had ERS activity



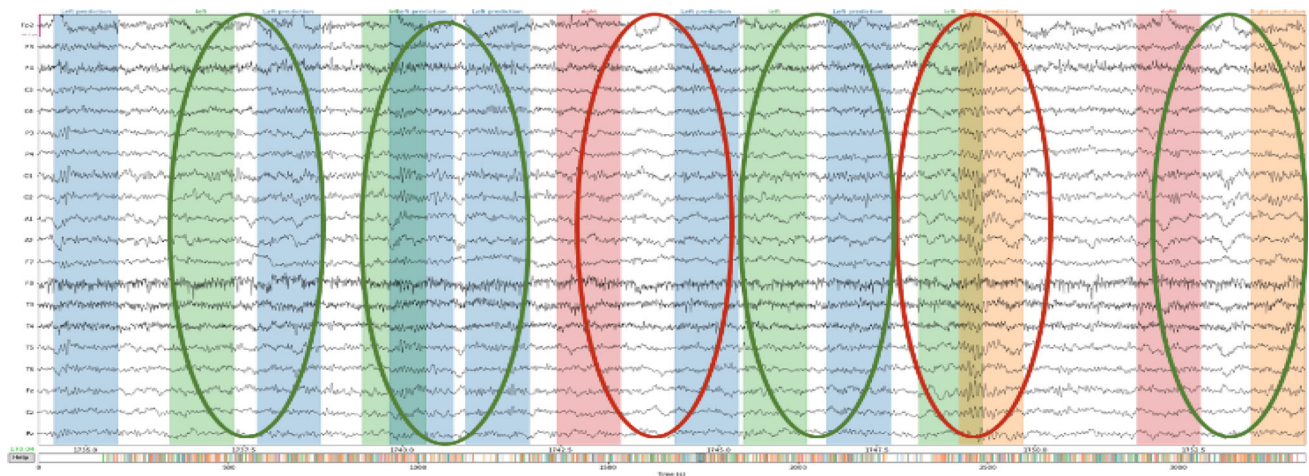


Fig. 8 Predictions on an unseen session. Almost all the classes were correctly classified (green circles), and the resting state was successfully detected. However, there is one right imagery detected as left and one

left detected as right (red circles). Red band: right imagery indication. Green band: Left imagery indication. Orange band: Right prediction. Blue band: Left prediction

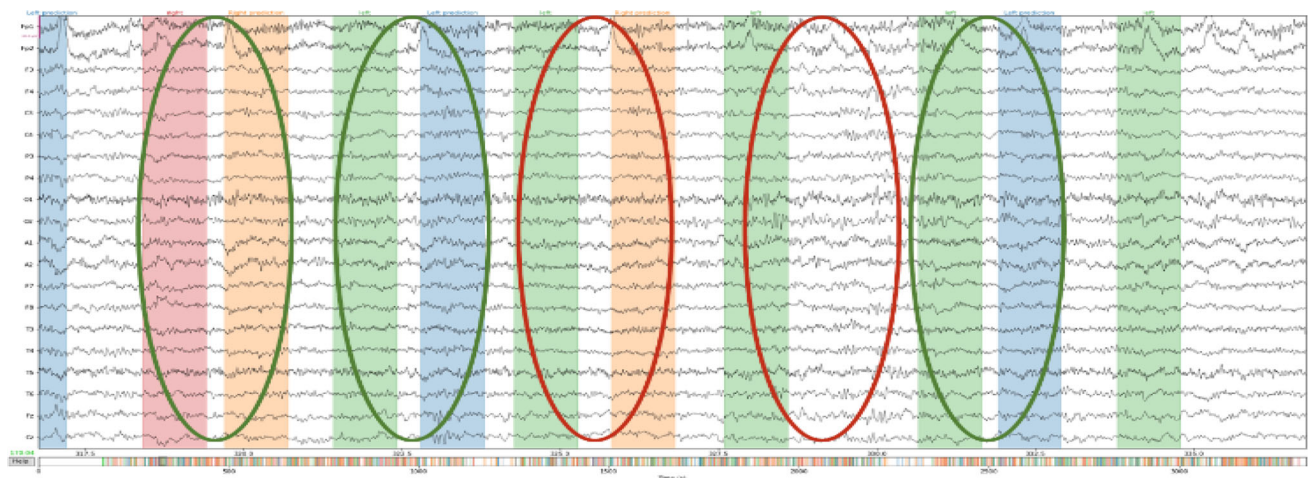


Fig. 9 Predictions on an unseen session. Almost all the classes were correctly classified (green circles). However, there is one left imagery detected as right and another left imagery that was not detected at all (red

circles). Red band: right imagery indication. Green band: Left imagery indication. Orange band: Right prediction. Blue band: Left prediction

in the Beta and not the Mu band. For the previously stated reasons, only the ERS from the Beta band was considered for the study. Only three participants (Subjects *C*, *E*, and *F*) proved to have good-quality recordings; hence, the other subjects were discarded for training the artificial intelligence models.

Before training the model, the data were preprocessed to enhance the overall performance. The ICA is the perfect solution to remove artifacts, such as eye-blinking, muscular or cardiac activity, without altering the crucial information of the signals. Furthermore, the data must be segmented correctly based on the duration of the imagery activity displayed in the ERD/ERS maps.

Moreover, a finite impulse response filter was carefully designed to maintain the Beta band's ERS for all the selected subjects (15–26 Hz). The filter's parameters were justified so that there are proper transition bands and no ripples in the passband. In addition, a Hamming window effectively increased the attenuation of the undesired secondary lobes that appear on the Bode. In addition, as the filter is intended for a real-time application, the order chosen was 150, so the delay is not noticeable (372.5 ms).

The next step was to train two models for classifying three classes: left imagery, right imagery, and a resting state. The hyperparameters of each architecture were chosen with the help of tuning and optimizing methods, which significantly enhanced the accuracy and precision.

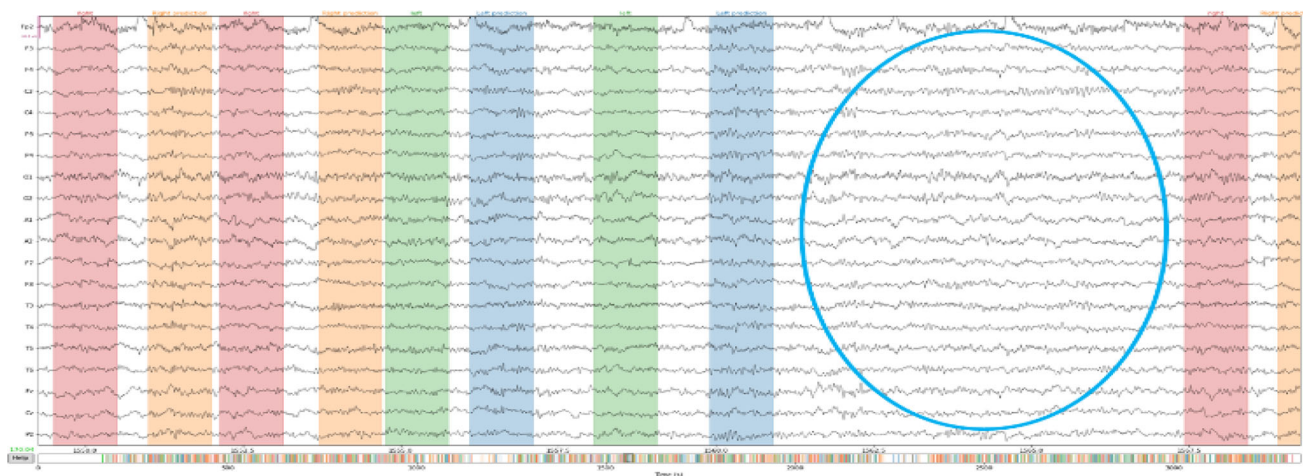


Fig. 10 Predictions on an unseen session. This case shows how well the model predicts a resting state, being passive when there is no indication to perform left or right-hand imageries (blue circle). Red band: Right

imagery indication. Green band: Left imagery indication. Orange band: Right prediction. Blue band: Left prediction

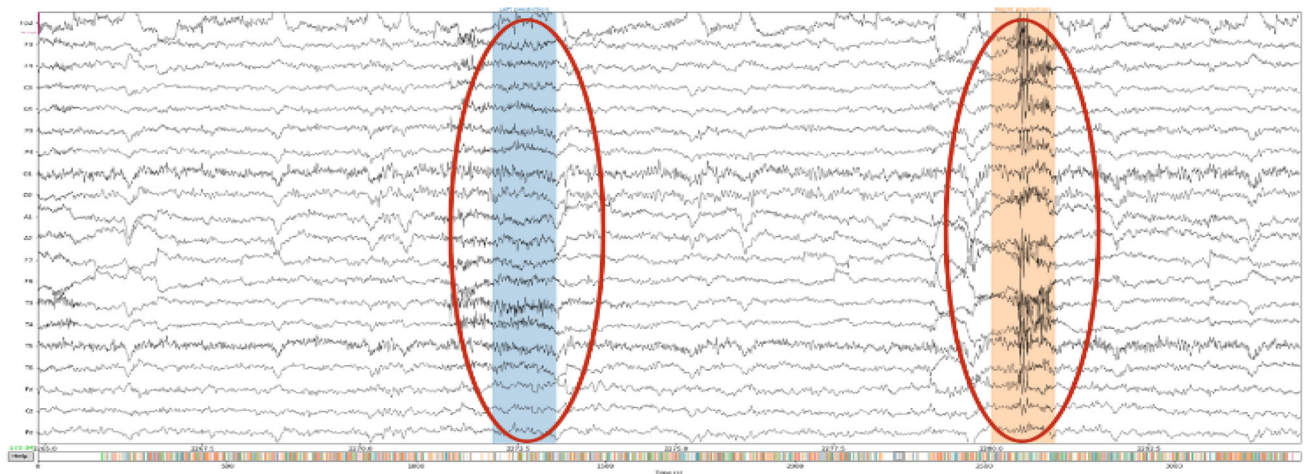


Fig. 11 Predictions on an unseen session. This case shows that sometimes an imagery state is detected even though the patient is in a resting state (red circles). Red band: Right imagery indication. Green band:

Left imagery indication. Orange band: Right prediction. Blue band: Left prediction

On the one hand, model LR, which classified left and right-hand imageries, was trained with various machine learning models with the CSP as features. The RF algorithm outperformed the other models with an overall accuracy of 83%, reaching 88% in Subject C; these great results demonstrate that the model generalizes the data and decently handles the inter- and intra-individual variability. The CSP undoubtedly does an excellent job discriminating between the left and right-hand classes.

On the other hand, model IR, which distinguishes between an imagery event and the resting state of the subject, has shown good performance with the STFT features in a CNN; the obtained accuracy was 78%. In addition, the precision was 90%, which is a good indicator that the model hardly has false

positives; this is important so that the robotic manipulator does not get triggered without any valid imagery from the user.

Both models worked cooperatively to do a real-time BCI. For testing this live analysis, an unseen session from Subject C was scrolled inside a time window of 0.9 s, constantly filtering and classifying the signals overlapping with the window. The results were outstanding and much more robust than expected: The overall system can detect the three classes with few erroneous classifications.

Our proposed method for distinguishing left and right-hand imagery is highly competitive with other related works in the literature, as evidenced by the accuracies reported in [30], which range from 80.5 to 85% for binary classification.



However, unlike those approaches, our system is able to accurately detect the resting state, which is critical for performing asynchronous MI-BCI. Other works such as [34, 35] do not achieve accurate and precise detection of the resting state, as the one obtained in this paper. Moreover, we did not find any other references where they present the implementation of an asynchronous MI-BCI in real time.

5 Conclusions

Our study resulted in the development of a highly accurate BCI system that successfully translates EEG signals into commands for controlling a robotic manipulator. By using preprocessing techniques and training two models, we achieved accurate classification of left and right-hand imageries, with an overall accuracy of 83%, reaching 88% in Subject C, and detection of the resting state, with an accuracy of 78% and a precision of 90%. These excellent results demonstrate the potential of our BCI system for practical applications, and its ability to detect the three classes with few erroneous classifications in real time.

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